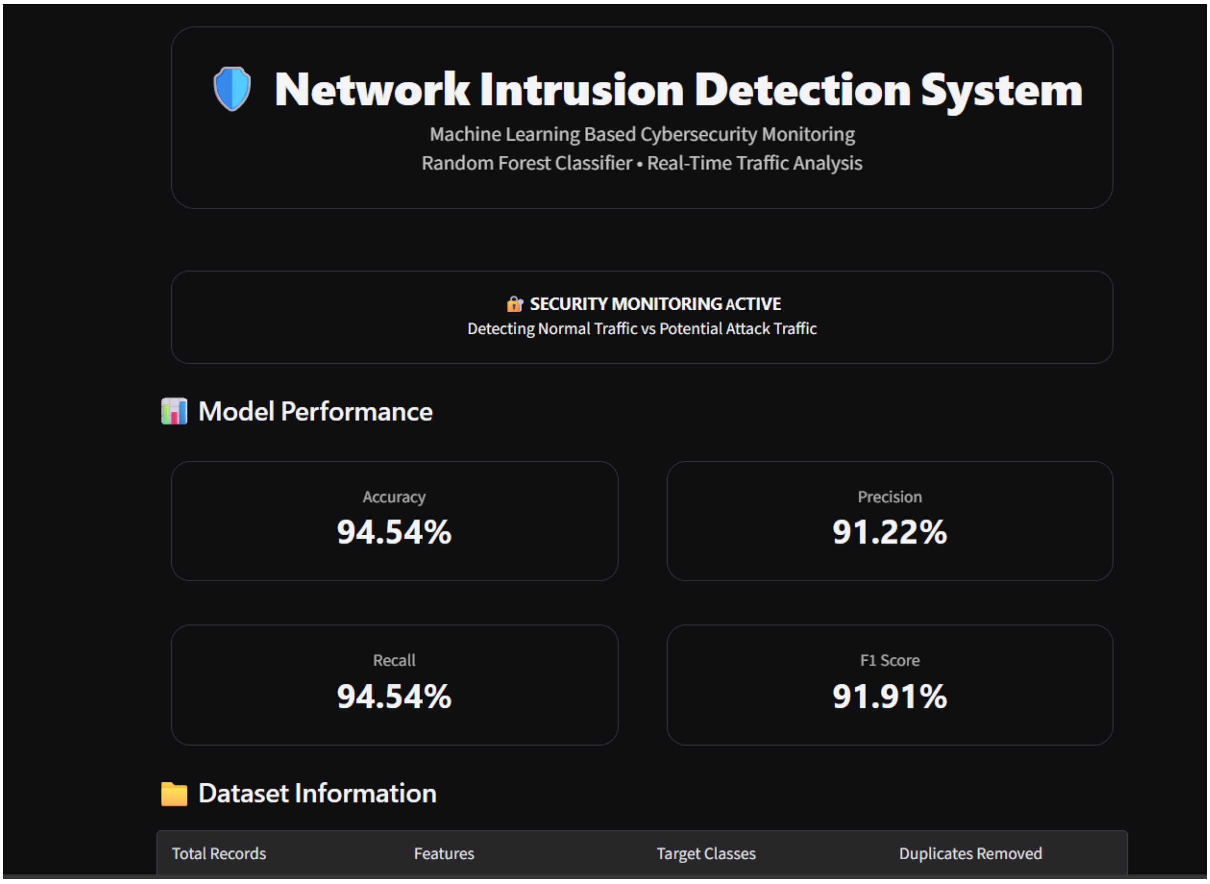
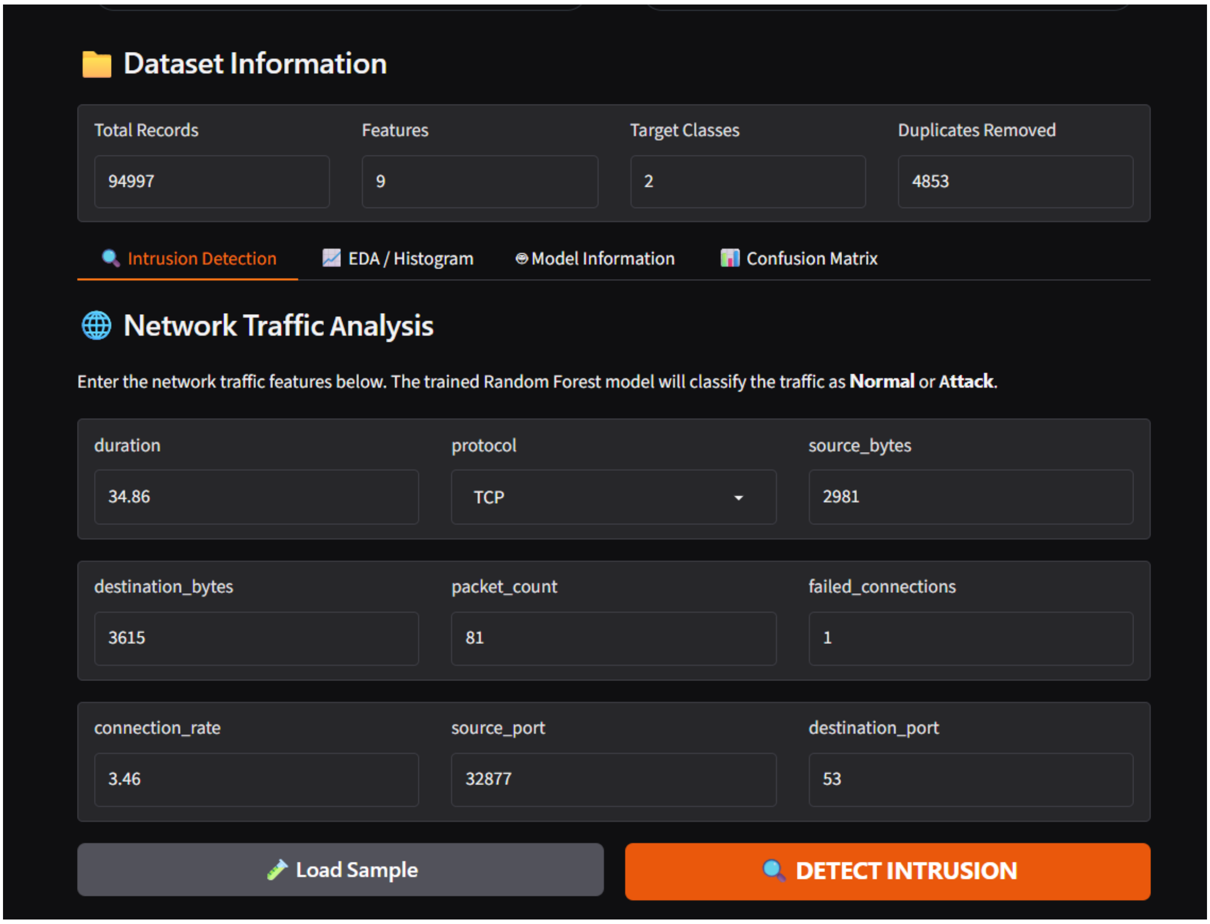
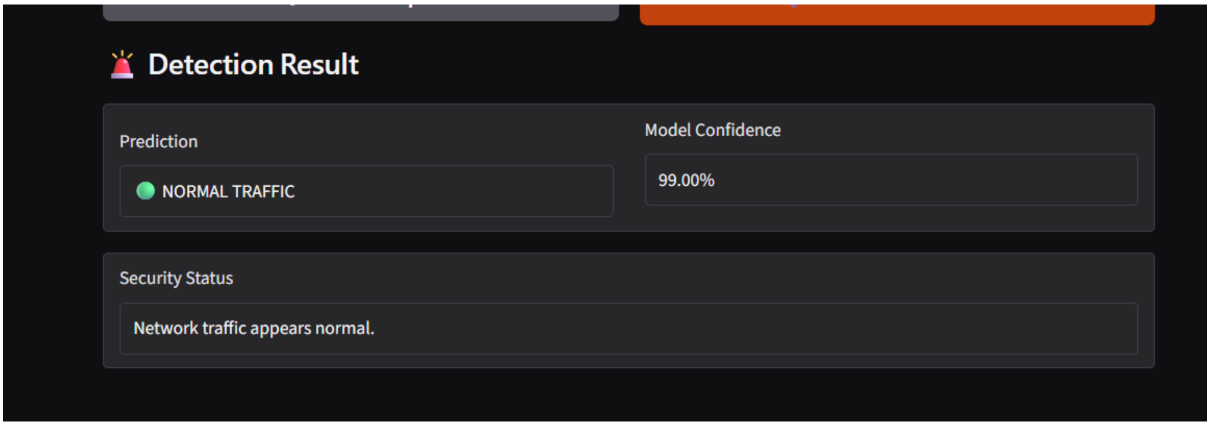
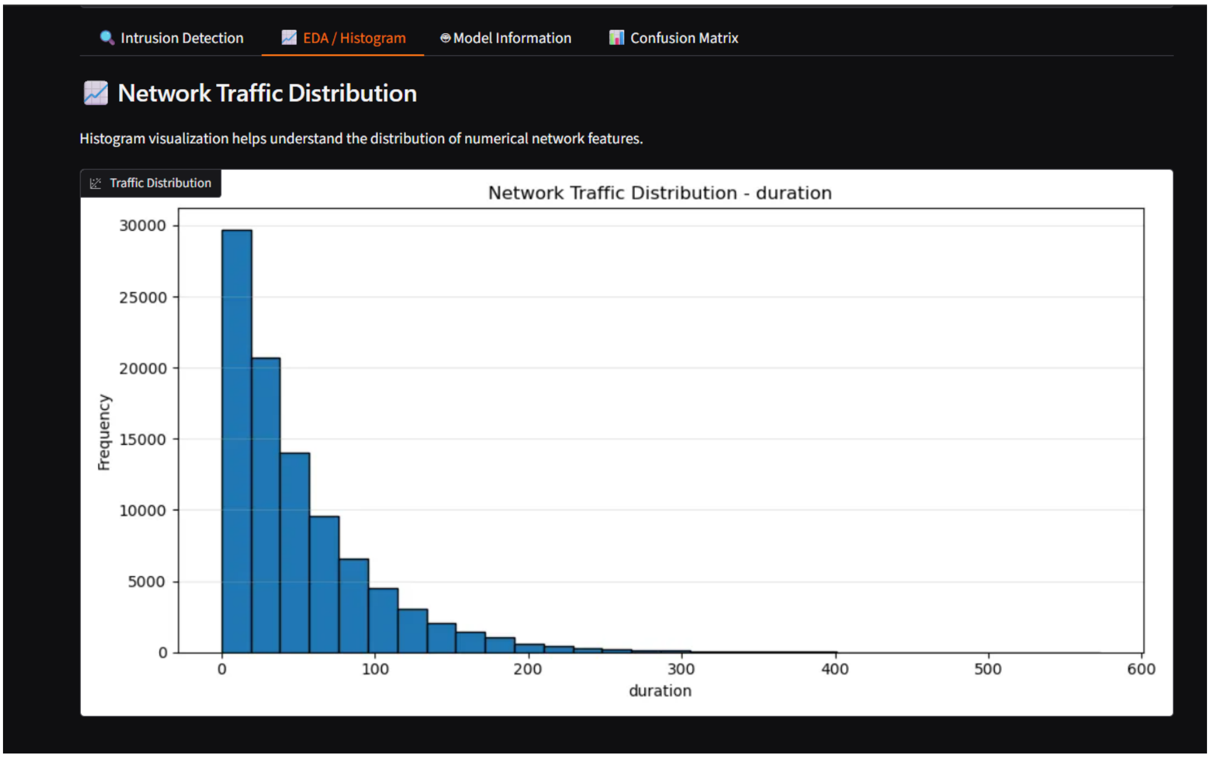

Network Intrusion Detection System
Machine Learning Based Cybersecurity Monitoring
Random Forest Classifier • Real-Time Traffic Analysis
SECURITY MONITORING ACTIVE
Detecting Normal Traffic vs Potential Attack Traffic
Model Performance
Accuracy
94.54%
Precision
91.22%
Recall
94.54%
F1 Score
91.91%
Dataset Information
Total Records
Features
Target Classes
Duplicates Removed


Dataset Information
Total Records
94997
Features
9
Target Classes
2
Duplicates Removed
4853
Intrusion Detection
EDA / Histogram
Model Information
Confusion Matrix
Network Traffic Analysis
Enter the network traffic features below. The trained Random Forest model will classify the traffic as Normal or Attack.
duration
34.86
protocol
TCP
source_bytes
2981
destination_bytes
3615
packet_count
81
failed_connections
1
connection_rate
3.46
source_port
32877
destination_port
53
Load Sample
DETECT INTRUSION


Detection Result
Prediction
NORMAL TRAFFIC
Model Confidence
99.00%
Security Status
Network traffic appears normal.


Intrusion Detection
EDA / Histogram
Model Information
Confusion Matrix
Network Traffic Distribution
Histogram visualization helps understand the distribution of numerical network features.
Traffic Distribution
Network Traffic Distribution - duration
Frequency
30000
25000
20000
15000
10000
5000
0
0
100
200
300
400
500
600
duration

📁 Dataset Information

Total Records	Features	Target Classes	Duplicates Removed
94997	9	2	4853

🔍 Intrusion Detection | 📈 EDA / Histogram | Model Information | 📊 Confusion Matrix

🤖 Random Forest Classifier

Algorithm: Random Forest

Number of Trees: 100

Class Weight: Balanced

Train/Test Split: 80% / 20%

Random State: 42

Preprocessing:

- Missing numerical values → Median
- Numerical scaling → StandardScaler
- Missing categorical values → Most Frequent
- Categorical encoding → OneHotEncoder

Target:

🟢 Normal Traffic

🔴 Attack Traffic

📋 Input Features

{ } Feature Names

```
[
  "duration",
  "protocol",
  "source_bytes",
  "destination_bytes",
  "packet_count",
  "failed_connections",
  "connection_rate",
  "source_port",
  "destination_port"
]
```

🛡 Network Intrusion Detection System
Powered by Python • Scikit-learn • Random Forest • Gradio

📁 Dataset Information

Total Records	Features	Target Classes	Duplicates Removed
94997	9	2	4853

🔍 Intrusion Detection 📈 EDA / Histogram ⚙ Model Information 📊 Confusion Matrix

📊 Confusion Matrix

The confusion matrix shows how accurately the model classified the test samples.

Confusion Matrix

0	1
2	1034
4	17960

🛡 Network Intrusion Detection System
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Machine Learning Cybersecurity Project